Math 243: Homework 01

Due Friday, Jan 24

Problem 1 (Arc length)**.** Let $f(x) = \frac{x^4}{8} + \frac{1}{4x^2}$. Compute the arc length of the curve from $x = 1$ to $x = 2$. Show your work.

Problem 2. Consider the following parametric equations

$$\begin{cases} x = 3\cos(t) + 4 \\ y = \sin(t) + 2 \end{cases}$$

for all t in the interval $[0, 2\pi]$.

(a.) Fill in the following table values, and plot the points on a graph.

t	$x(t)$	$y(t)$
0		
$\pi/6$		
$\pi/4$		
$\pi/3$		
$\pi/2$		
$2\pi/3$		
$3\pi/4$		
$5\pi/6$		
π		

(b.) Convert the parametric equations to Cartesian form by eliminating the variable t. *Hint: use the trig identity* $\sin^2(t) + \cos^2(t) = 1$

(c.) Sketch the graph of the curve as t ranges from 0 to 2π.